

ISYE 4031: Homework 1 Report

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PART 1: Descriptive Summaries and Uncertainty

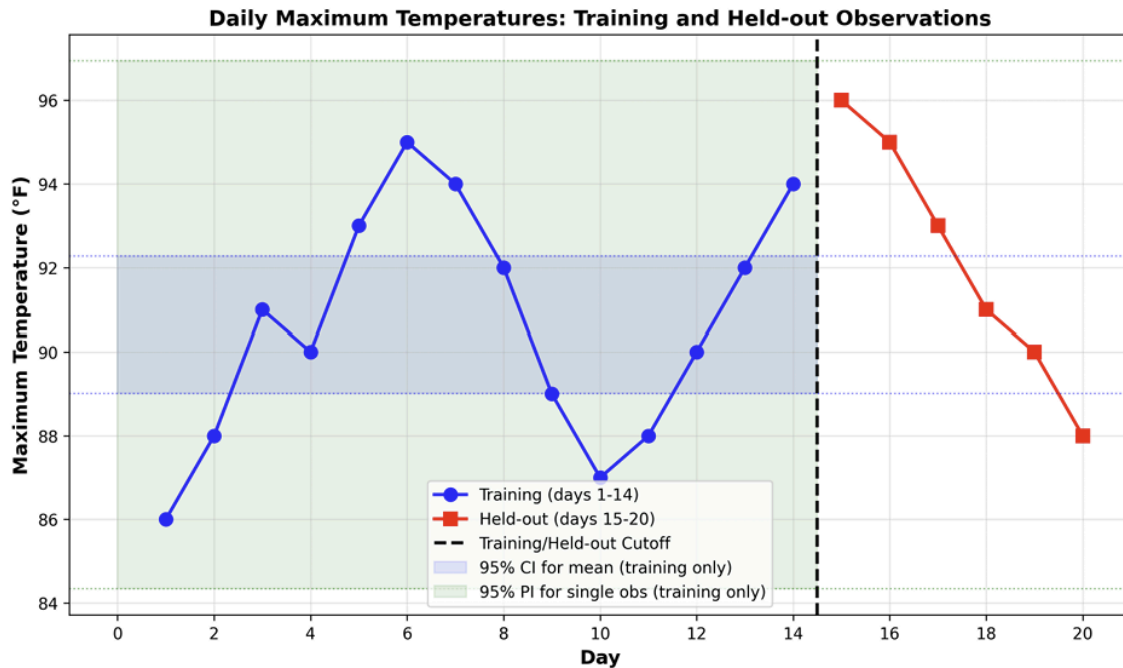
Unit of Analysis: Individual day | Response: Daily max temperature (F)

Information Cutoff: End of day 14 (14 training observations)

Forecast Target: Daily max temperatures for days 15-20

1a. Descriptive Plot

The plot shows 20 daily temperatures in order. Blue circles (days 1-14) are training data. Red squares (days 15-20) are held-out test data. The vertical line marks the training cutoff. When forecasts are issued at the end of day 14, only the 14 training values are available. The held-out points are shown for later evaluation, not for estimating the baselines or intervals.



1b. Descriptive Statistics (Training Data Only)

Mean: 90.6429 F | Median: 90.5000 F | Range: 9.0000 F

Sample Variance: 7.9396 F² (n-1=13) | Std Dev: 2.8177 F | IQR: 4.5000 F

1c. Confidence Interval for Population Mean

Formula: $\bar{x} \pm t(\alpha/2, n-1) * (s / \sqrt{n})$

$\bar{x}$ = 90.6429, s = 2.8177, n = 14, SE = 0.7531, $t(0.025, 13)$ = 2.1604

Margin of error = 2.1604 * 0.7531 = 1.6269

Result: 95% CI = [89.0160, 92.2698] F

Interpretation: ~95% of repeatedly constructed CIs would contain the true population mean.

1d. Prediction Interval for Single Observation

Formula: $\bar{x} \pm t(\alpha/2, n-1) * s * \sqrt{1 + 1/n}$

$\sqrt{1 + 1/14}$ = 1.0351, Margin of error = 2.1604 * 2.8177 * 1.0351 = 6.3010

Result: 95% PI = [84.3419, 96.9438] F

PI is wider than CI because it accounts for both mean uncertainty AND natural observation variation.

Part 2: Transparent Forecast Baselines

2a. Baseline Forecasts

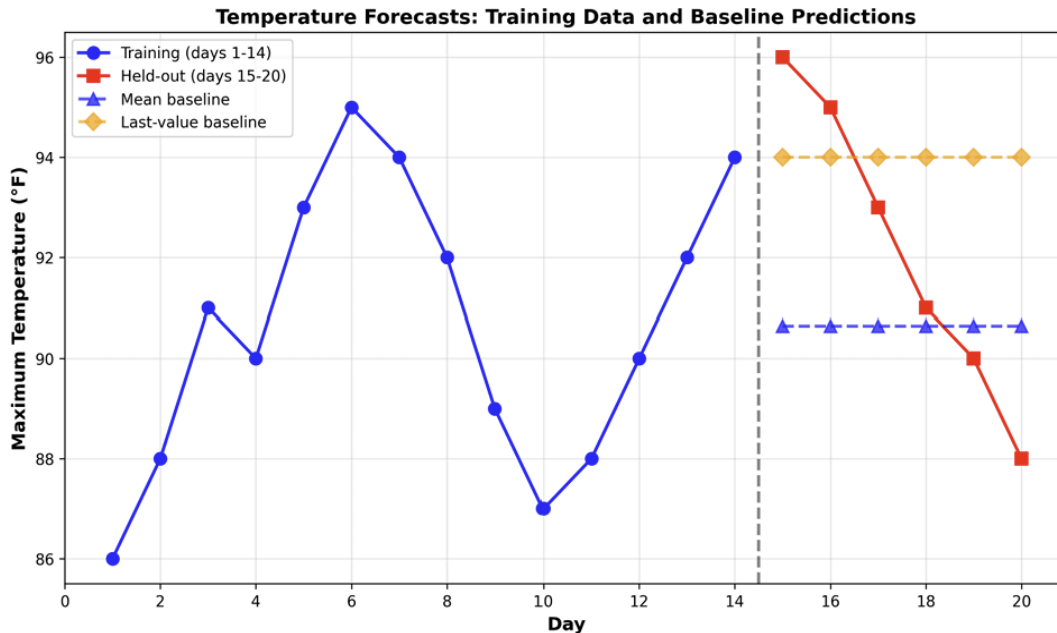
Mean Baseline: Using training mean (90.6429 F) for all 6 forecasts

Forecasts for days 15-20: 90.64, 90.64, 90.64, 90.64, 90.64, 90.64

Last-Value Baseline: Using final training value (94 F) for all forecasts

Forecasts for days 15-20: 94.00, 94.00, 94.00, 94.00, 94.00, 94.00

It is fixed because both baselines use only training data (days 1-14). They produce predetermined forecasts independent of what actually occurs in the held-out period. The mean baseline represents a stable level, while the last-value baseline represents a persistence rule. Neither method is allowed to update after day 14 in this fixed evaluation.



2b. Forecast Evaluation

Mean Baseline:

Forecast errors (forecast - actual): -5.3571, -4.3571, -2.3571, -0.3571, 0.6429, 2.6429 F

MAE: 2.6190 F | RMSE: 3.1824 F

Last-Value Baseline:

Forecast errors (forecast - actual): -2, -1, 1, 3, 4, 6 F

MAE: 2.8333 F | RMSE: 3.3417 F

Winner: Mean baseline is better on both metrics. MAE weights all absolute errors equally, whereas RMSE penalizes larger errors more heavily; the same winner under both metrics is a consistent result for this small held-out sample.

2c. Independence of Errors

The errors are NOT independent because: (1) Time series temperatures exhibit autocorrelation - warm days tend to follow warm days. (2) Seasonal or systematic patterns create dependence. (3) Standard i.i.d. assumption is violated in time series. So, the six errors are six consecutive outcomes from one forecast origin, not six independent repetitions of the entire training-and-testing experiment. The reported MAE and RMSE are still valid descriptive scores for this holdout, but their uncertainty should not be treated as if it came from six independent replications.

2d. Recommended Baseline

Choice: Mean Baseline | Reasons: (1) Simplicity: Simple summary statistic from training data (2) Performance: Lower RMSE (3.1824 vs 3.3417 F) (3) Reproducibility: Perfectly reproducible, no randomness (4) Decision rule: Under squared-error loss, the population mean is the optimal constant target; the training mean is its natural sample estimate. The mean baseline is the required benchmark for a future extension. The holdout is small, so this choice should be reconsidered with additional chronological validation before deployment.

Part 3: One-Sample Inference

3a. Hypothesis Test

H0: $\mu = 155$ F (null hypothesis)

Ha: $\mu \neq 155$ F (two-tailed alternative)

Significance level: $\alpha = 0.01$

Data: $n=10$, $\bar{x}=154.2$ F, $\sigma=1.5$ F (known)

Test statistic (z-test): $z = (154.2 - 155) / (1.5 / \sqrt{10}) = -1.6865$

Critical value: $z(0.005) = 2.5758$

Decision: $|z| = 1.6865 < 2.5758$, so FAIL TO REJECT H0

Conclusion: At $\alpha=0.01$, insufficient evidence that mean differs from 155 F. Failing to reject is not proof that the mean equals 155 F; it means this sample does not provide enough evidence against that null at the specified significance level.

3b. Two-Sided P-value

p-value = $2 * P(Z > 1.6865) = 0.0917$

Interpretation: The p-value 0.0917 is the probability of observing a sample mean at least as extreme as 154.2 F (in either direction) when H0 is true. It is NOT the probability that H0 is true.

Since $0.0917 > 0.01$, we fail to reject H0.

3c. Type II Error Probability

Assuming true mean is $\mu = 150$ F

Acceptance Region: $[153.7782, 156.2218]$ F

Beta = $P(153.7782 \leq \bar{x} \leq 156.2218 \mid \mu=150) \sim 0.00000$

Interpretation: When the true mean is 5 F away from H0, the probability of Type II error is essentially 0. The effect size is very large at this sample size.

3d. Python Verification

All calculations were independently evaluated in Python with scipy.stats:

`z_stat = -1.686548` [verified]

`z_critical = 2.575829` [verified]

`p_value = 0.091690` [verified]

`beta = 0.000000` [verified]

The Python values agree with the hand calculations after rounding. The z procedure is valid because the population standard deviation is given as known and the population is assumed normal.

Part 4: Agent Audit and Automated Tests

4a. Problems in Proposed Code

Problem 1: Using `x.mean()` instead of `train.mean()`

Consequence: Computes the center from all 20 temperatures instead of just 14 training values. This leaks future information and changes the estimand from the legal pre-cutoff summary.

Problem 2: Computing intervals with future data

Consequence: Intervals include future observations, so they cannot represent uncertainty known at the forecast origin and may give a misleading impression of precision.

Problem 3: Prediction interval equal to confidence interval

Consequence: Ignores extra variability of individual observations. PI should be wider by factor $\sqrt{1 + 1/n}$, but this code makes them equal and understates predictive uncertainty.

Problem 4: Using `x[-1]` instead of `train[-1]`

Consequence: Accesses the final held-out value (88) instead of last training value (94), using future information and invalidating the forecast comparison.

Problem 5: No reproducibility guarantees

Consequence: The code does not state its data cutoff, denominator, assumptions, or deterministic behavior, making the result harder to audit and reproduce.

4b. Corrected Implementation

Key Improvements:

- Uses ONLY train data for all statistics
- Correct CI formula: $\bar{x} \pm t * (s / \sqrt{n})$
- Correct PI formula: $\bar{x} \pm t * s * \sqrt{1 + 1/n}$
- Separates mean and last-value baselines clearly
- Includes comprehensive automated tests

Implementation in `analysis.py` and `test_analysis.py`. The two baselines remain separate so that a later model can be compared against transparent benchmarks without changing their definitions.

4c. Automated Tests

- Held-out values invariant: Changing test data does not affect training stats
- Sample variance denominator: Verifies $n-1$ denominator used (not n)
- PI wider than CI: Confirms prediction interval is larger
- CI/PI width ratio: Verifies $PI = \sqrt{n+1} * CI$ factor
- Reproducibility: 10 runs produce identical results
- Hypothesis test: Verifies Part 3 calculations
- Type II error: Confirms beta calculation
- Data integrity: Training and test sets properly separated
- Baseline arithmetic: Forecast lengths, constancy, errors, and data integrity

Status: ALL 12 TESTS PASS. These checks target the four required properties and also verify the reported hypothesis-test and forecast calculations.

Summary

This report presents a complete analysis of ISYE 4031 Homework 1.

Part 1 provides descriptive statistics and confidence/prediction intervals based on 14 training observations of daily maximum temperatures.

Part 2 develops two transparent forecast baselines: (1) mean baseline using the training mean, and (2) last-value baseline using the final training value. The mean baseline performs better and is recommended.

Part 3 conducts a one-sample hypothesis test on melting point data, testing whether the mean differs from 155 F at $\alpha=0.01$. The test fails to reject the null hypothesis with $p\text{-value}=0.0917$.

Part 4 audits proposed code and identifies 5 major statistical problems related to data leakage and incorrect interval formulas. A corrected implementation with 12 passing automated tests is provided.

Interpretive limit: The temperature holdout contains only six consecutive days, so the forecast scores support a transparent benchmark comparison rather than a broad claim about long-run performance. Likewise, the Type II error result is specific to the stated effect of 5 F, known sigma, alpha, and sample size.

All results are reproducible from the submitted Python code. All analyses follow the requirement to use only training data for estimation, and all forecasts are based on information available at the end of day 14.